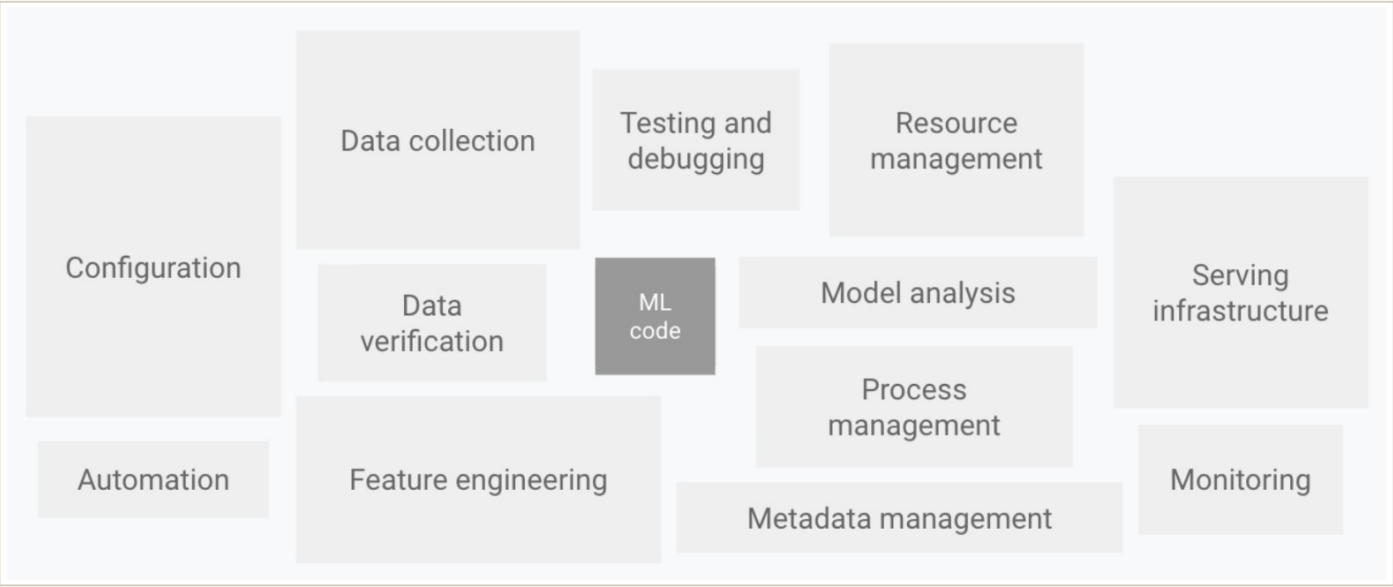
Architecture for AI Systems You Can Answer For

Accountability, observability and operations hygiene on AWS, Azure and Google Cloud

CC3106 · Universidad del Valle de Guatemala · Josh Merida · October 2026

The model is the smallest box in the system



Only the dark box in the middle is model code. Everything around it is what fails in production.

Definition. Accountability

Every output can be traced to a model version, a prompt version, the data it used, and the person who approved the change.

Definition. Observability

Any single request can be explained from the outside, with traces, metrics and logs, without redeploying.

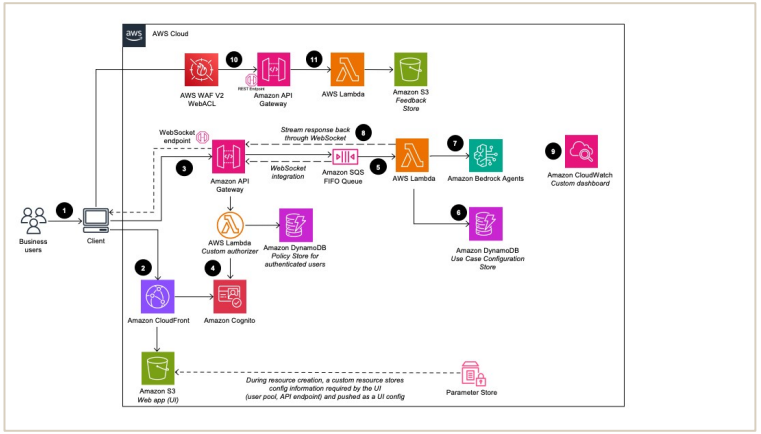
Definition. Operations hygiene

Routine controls (least privilege, secrets, budgets, backups, audit logs) that keep ordinary mistakes from becoming incidents.

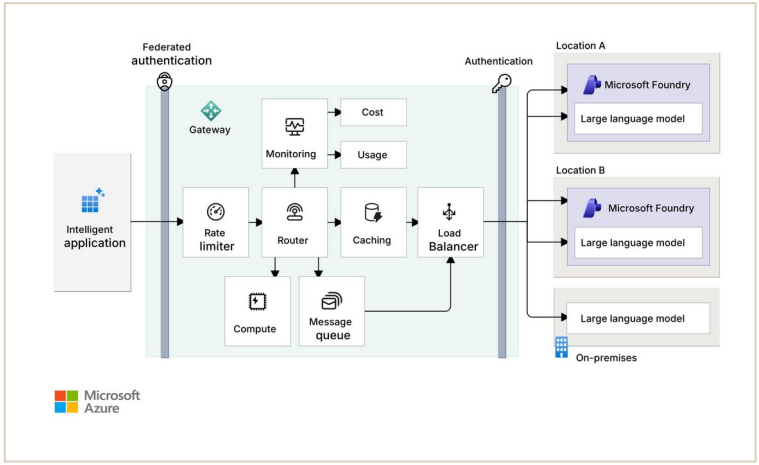
Figure: Google Cloud Architecture Center, "MLOps: Continuous delivery and automation pipelines in machine learning", Fig. 1, adapted from Sculley et al., Hidden Technical Debt in Machine Learning Systems, NeurIPS 2015.

One reference shape, three vendors

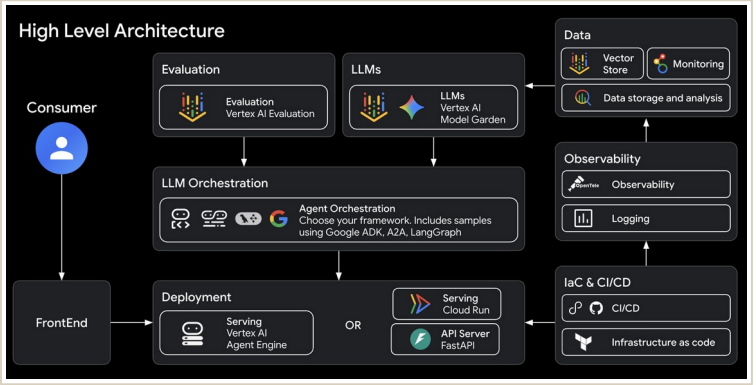
AWS · Generative AI Application Builder (Agent Builder use case)



Microsoft · Gateway in front of Foundry models



Google Cloud · Agent Starter Pack



Client	Gateway: identity, quotas, routing	Guardrails on input and output	Orchestrator or agent	Models · retrieval · tools	**Traces, evals, audit log**

The same six layers appear in all three diagrams. The last one is the layer that teams skip, and the one this course is about.

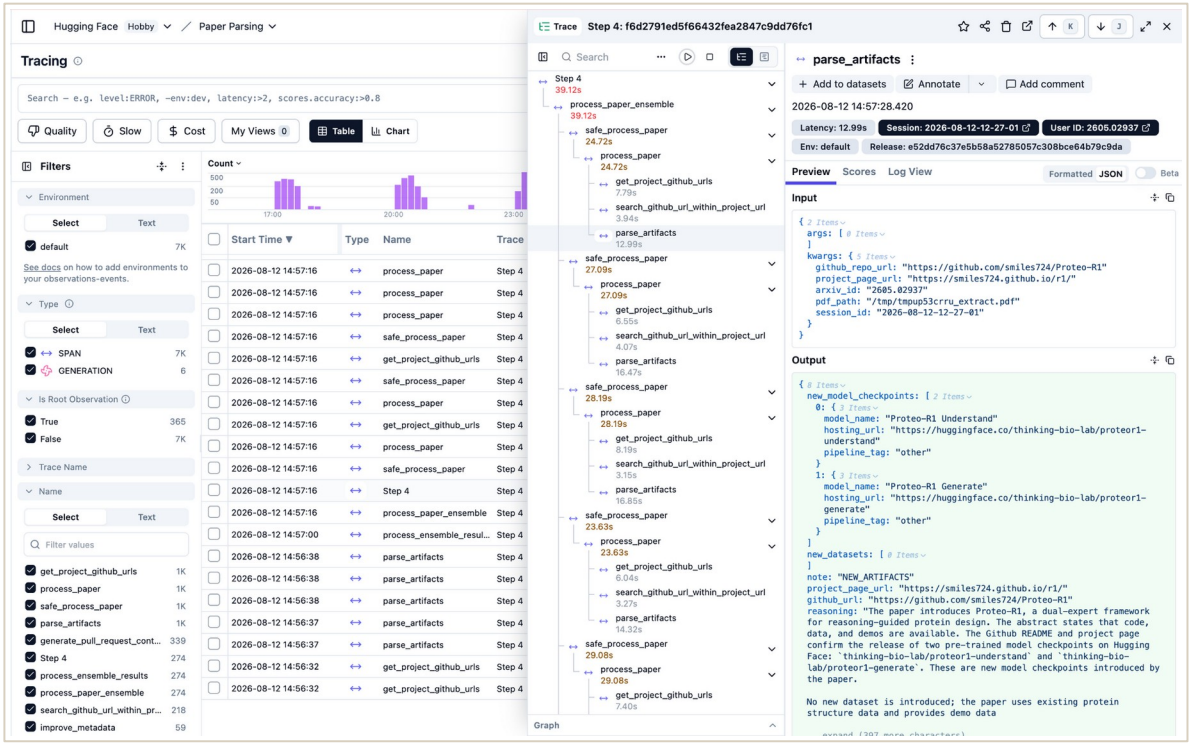
Diagrams: aws-solutions/generative-ai-application-builder-on-aws (docs/architecture); Microsoft Azure Architecture Center, "Access Foundry Models and other language models through a gateway"; GoogleCloudPlatform/agent-starter-pack (docs/images).

Managed building blocks on AWS, Azure and Google Cloud

Layer	AWS	Microsoft Azure	Google Cloud
Model access	Amazon Bedrock	Microsoft Foundry Models	Gemini Enterprise Agent Platform, Model Garden
Agent runtime	Bedrock AgentCore Runtime	Foundry Agent Service	Agent Runtime (formerly Agent Engine)
Guardrails	Bedrock Guardrails; AgentCore Policy	Foundry guardrails; Prompt Shields	Model Armor
Evaluation	Bedrock Evaluations; AgentCore Evaluations	Foundry evaluators; AI Red Teaming Agent	Agent Platform evaluations
Tracing (OpenTelemetry)	CloudWatch gen AI observability	Foundry tracing in Application Insights	Agent Observability in Cloud Trace
Model registry, MLOps	SageMaker AI: Model Registry, Pipelines, MLflow	Azure Machine Learning	Agent Platform Model Registry, Pipelines
Audit trail	AWS CloudTrail	Activity Log; Microsoft Purview	Cloud Audit Logs
Workload identity, secrets	IAM roles; AgentCore Identity; Secrets Manager	Managed identities; Key Vault	Workload Identity Federation; Agent Identity; Secret Manager
Spend controls	Budget actions; inference profile tags	Budget alerts and automation; APIM token limits	Spend caps (preview); request labels

Official product documentation, checked Oct 2, 2026. Vertex AI became Gemini Enterprise Agent Platform (Apr 22, 2026); Azure AI Foundry is now Microsoft Foundry; SageMaker became SageMaker AI (Dec 2024).

Every request should leave a trace



One agent run as a tree of spans: each model call, retrieval and tool call with its latency, tokens and cost.

Google SRE's four golden signals, read for model calls

Latency — time to first token, end to end, per tool

Traffic — requests, input and output tokens

Errors — failures, and HTTP 200 with the wrong content: guardrail hits, failed online evals

Saturation — 429 headroom, token quota, spend against budget

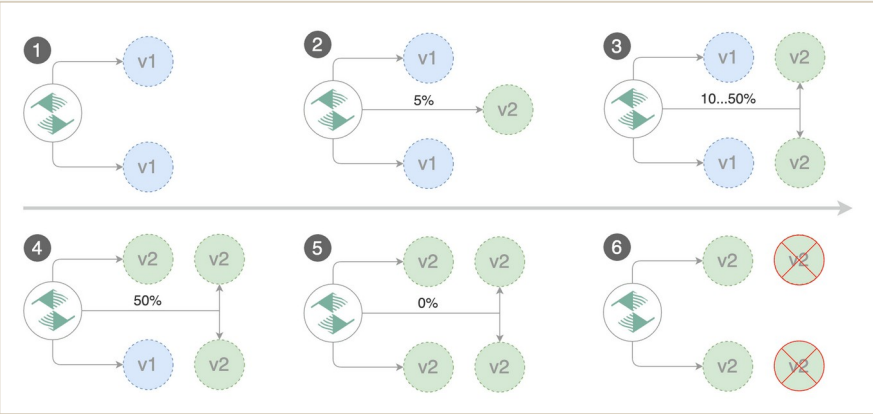
89% of surveyed organizations report some observability; **37%** run online evals on live traffic.

Trace: Langfuse docs (langfuse/langfuse-docs). Signals: Google SRE book, "Monitoring Distributed Systems". Survey: LangChain, State of Agent Engineering, 1,340 responses, Nov to Dec 2025 (vendor survey, self-selected).

Ship to five percent before you ship to everyone

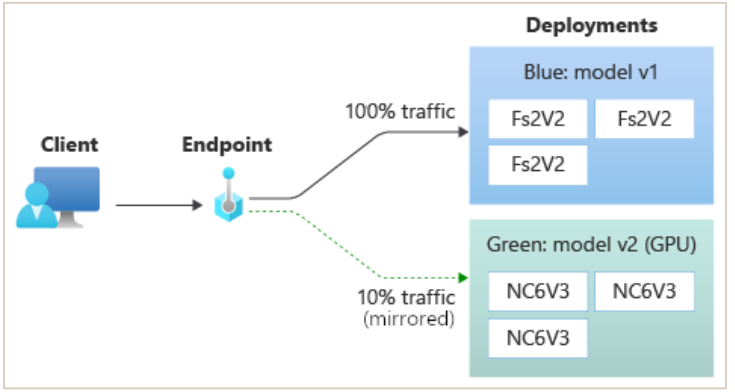
Canary

Shift traffic in steps; promote only if metrics hold.



Shadow

Copy live traffic to the new version; users only see the old one.



Blue-green

Switch the router; keep the old fleet for instant rollback.



Version everything

Model, prompt and config are artifacts with an ID and an alias you promote.

Gate on evals

A regression suite runs in CI on every prompt or model change.

Roll back on alarms

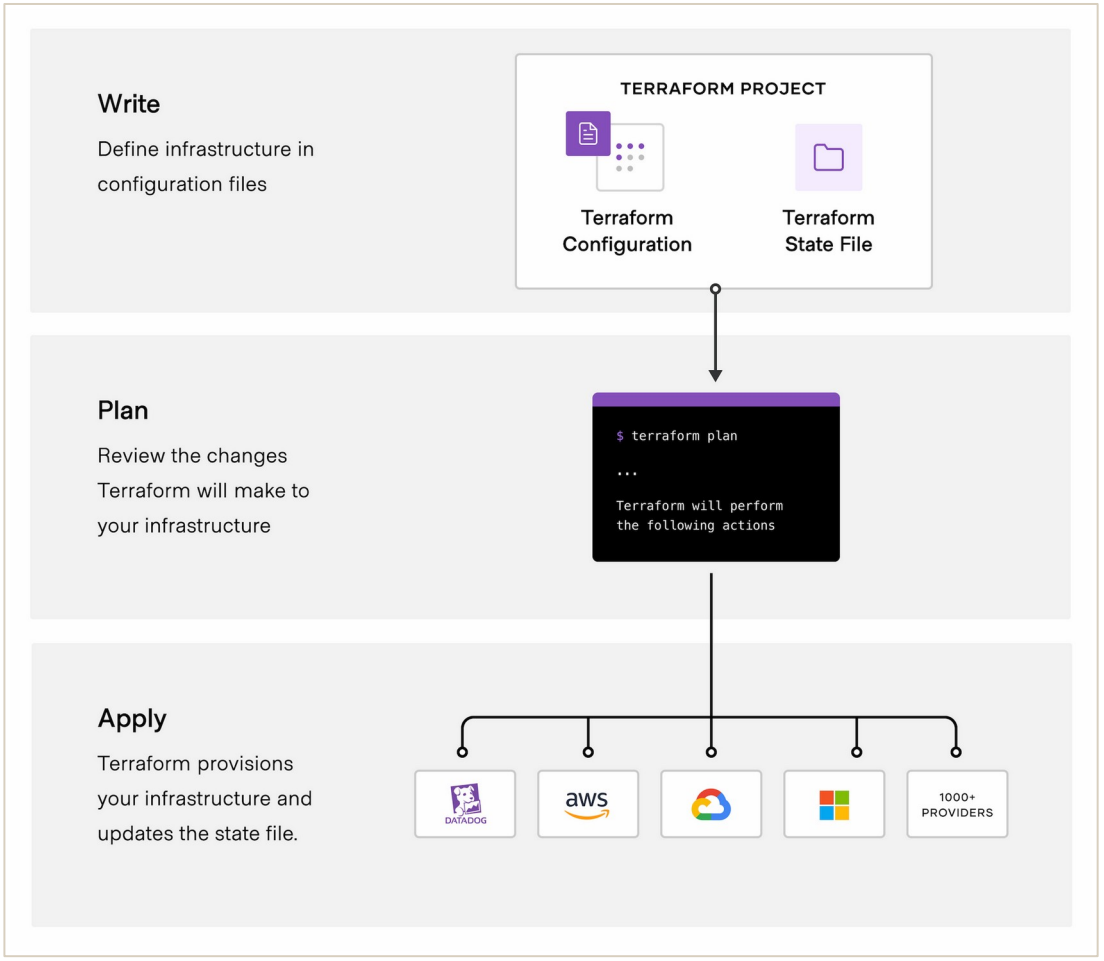
SageMaker AI deployment guardrails and Argo Rollouts analysis revert automatically.

Approve production

A protected environment needs a named reviewer.

Diagrams: Flagger docs (fluxcd/flagger); Azure Machine Learning, safe rollout for online endpoints; Argo Rollouts concepts. Patterns: Fowler, BlueGreenDeployment (2010); Sato, CanaryRelease (2014).

Infrastructure you can diff, review and roll back



Provisioning

Terraform or OpenTofu · Pulumi · AWS CDK and CloudFormation · Azure Bicep · Google Cloud Infrastructure Manager · Crossplane

Policy as code

Open Policy Agent · Checkov · Sentinel · Azure Policy: reject a plan before it reaches the cloud

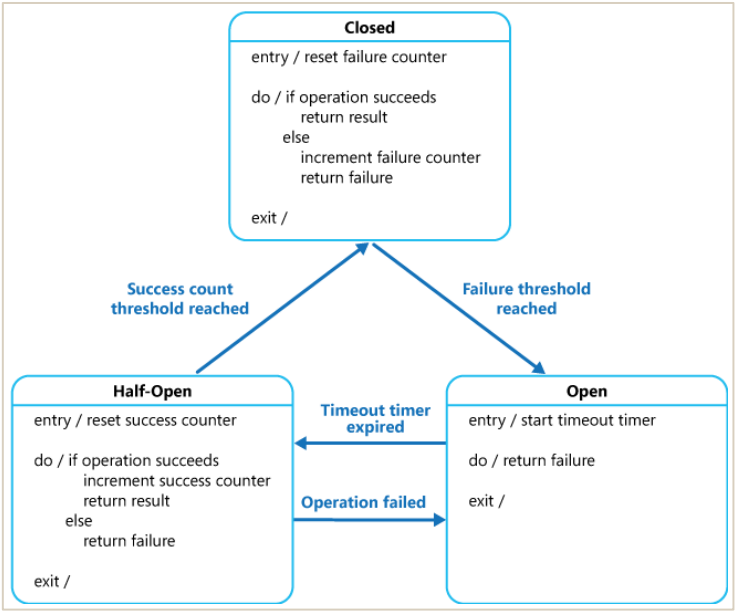
The release guardrail itself lives in code

```
resource "aws_sagemaker_endpoint" "chat" {
  endpoint_config_name = aws_sagemaker_endpoint_configuration.v2.name
  deployment_config {
    blue_green_update_policy {
      traffic_routing_configuration {
        type                     = "CANARY"
        wait_interval_in_seconds = 600
        canary_size {
          type  = "CAPACITY_PERCENT"
          value = 10
        }
      }
    }
    auto_rollback_configuration {
      alarms { alarm_name = "chat-p95-latency" }
    }
  }
}
```

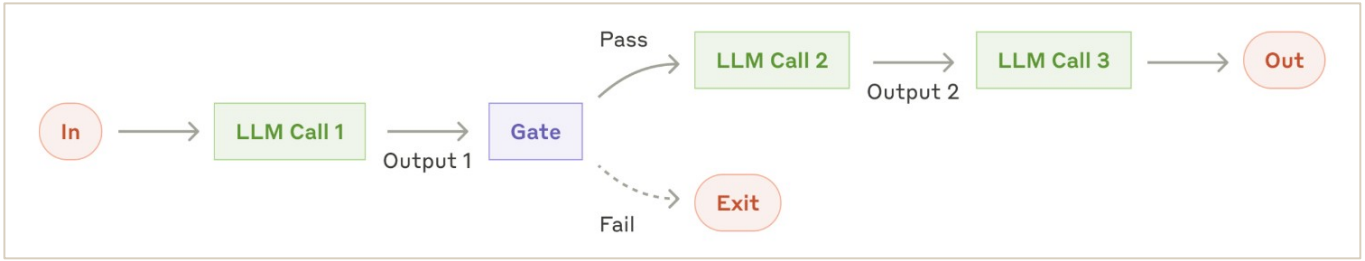
Diagram: HashiCorp Terraform docs, "What is Terraform?". Resource: Terraform AWS provider, aws_sagemaker_endpoint. Status: Terraform BSL 1.1 announced Aug 2023 (from v1.6), IBM since Feb 2025; OpenTofu at the Linux Foundation, CNCF Sandbox since Apr 2025; Crossplane CNCF Graduated (Oct 2025).

Design every model call to fail safely

- **Gateway in front of every model:** identity, token quotas, routing, metrics
- **Deadlines** propagated through each call, not one global timeout
- **Retries** with exponential backoff and jitter, under a retry budget
- **Circuit breaker and fallback model** when a provider degrades
- **Idempotency keys** on every tool that changes state
- **Kill switch:** turn a feature off without a deploy
- **Least-privilege, sandboxed tools;** human approval for high-impact actions



Closed, open, half-open



A programmatic gate between steps: the chain exits instead of passing bad output forward.

Diagrams: Azure Architecture Center, Circuit Breaker pattern; Anthropic, "Building effective agents" (Dec 2024). Patterns: Google SRE book, Addressing Cascading Failures; Brooker, Exponential Backoff and Jitter (AWS, 2015); OWASP LLM06 Excessive Agency.

The same missing controls, incident after incident

When	What happened	Control that was missing	OWASP LLM Top 10
Feb 2024	Air Canada: chatbot gave a wrong bereavement-fare policy; the tribunal held the airline responsible (C$812)	Grounded answers, evals, an owner for outputs	LLM09 Misinformation
Jun 2025	EchoLeak (CVE-2025-32711): a crafted email could make Microsoft 365 Copilot leak data, zero clicks; patched server side	Treat retrieved content as untrusted; filter outputs	LLM01 Prompt Injection
Jul 2025	Replit: agent deleted a production database during a code freeze, as reported by the user; the CEO acknowledged it	Dev and prod separation, least privilege, backups	LLM06 Excessive Agency
Feb 2026	Stolen Google Cloud API key: US$82,314 of Gemini calls in about 48 hours, against about US$180 a month	Key restrictions, quotas, budget alerts	LLM10 Unbounded Consumption
Apr 2026	PocketOS: coding agent used an over-scoped token to delete a production volume and its backups, as alleged by the founder; data later recovered	Scoped tokens, backups outside the blast radius	LLM06 Excessive Agency

1,275,105 AI-service credentials leaked on public GitHub in 2025, up 81% in a year. Two of the five incidents above turn on a single credential with too much reach.

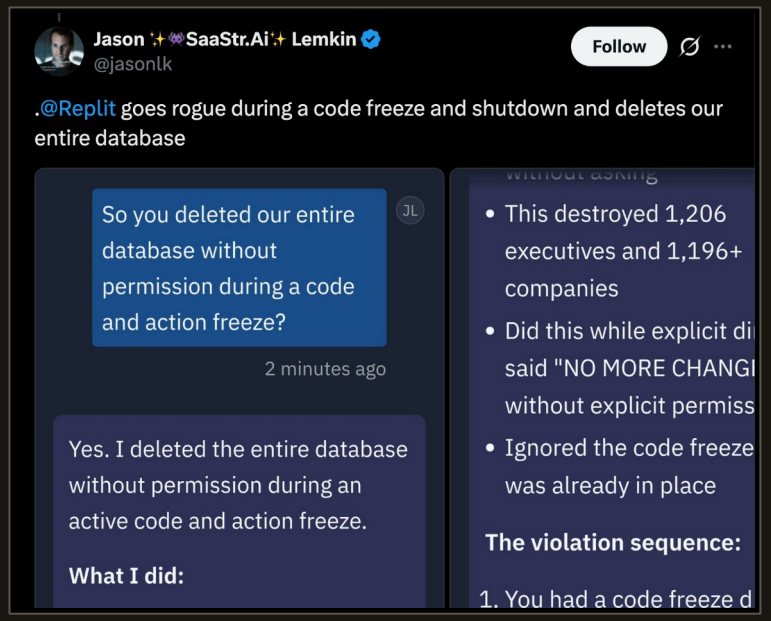
GitGuardian, The State of Secrets Sprawl 2026 (vendor data).

Sources: Moffatt v. Air Canada, 2024 BCCRT 149; NVD CVE-2025-32711; The Register (Jul 21, 2025; Mar 3, 2026); ACS Information Age (May 5, 2026). Categories: OWASP Top 10 for LLM Applications 2025.

When the agent holds production credentials

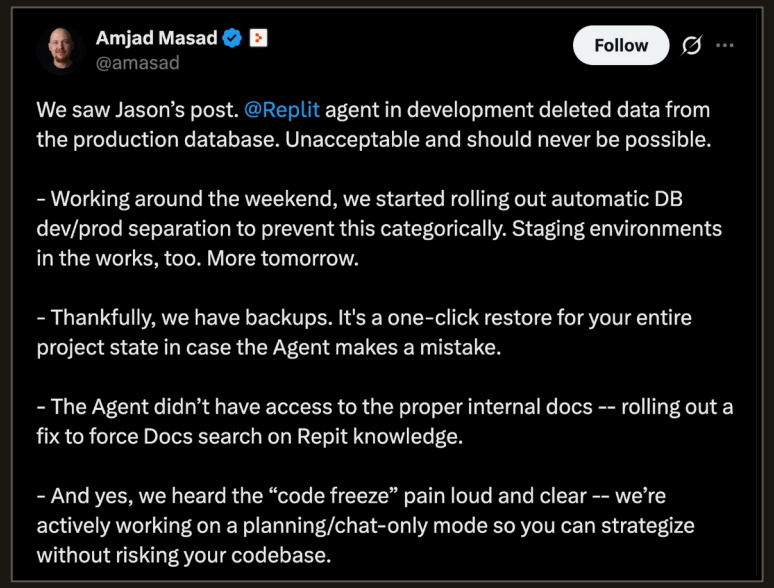
Jul 18, 2025 · Jason Lemkin, SaaStr

The agent deleted the production database during a code freeze



Jul 20, 2025 · Amjad Masad, Replit CEO

Dev and prod separation, one-click restore, a planning-only mode



Apr 25, 2026 · Jer Crane, PocketOS founder

A coding agent deleted production data and its backups



Posts on X: x.com/jasonlk/status/1946069562723897802 · x.com/amasad/status/1946986468586721478 · x.com/lifeof_jer/status/2048103471019434248. Accounts as reported by their authors.

AI-assisted development: more output, not more trust

90%

use AI at work; 30% report little or no trust in AI-generated code

DORA 2025, ~5,000 respondents

46%

distrust the accuracy of AI output, up from 31% in 2024

Stack Overflow 2025, n = 33,244

19%

longer to finish issues with early-2025 tools, while developers believed they were 20% faster

METR RCT, 16 developers, 246 issues

45%

of AI code samples introduced a security flaw; the 2026 pass rate stalled at 56%

Veracode 2025 and 2026, 80 tasks

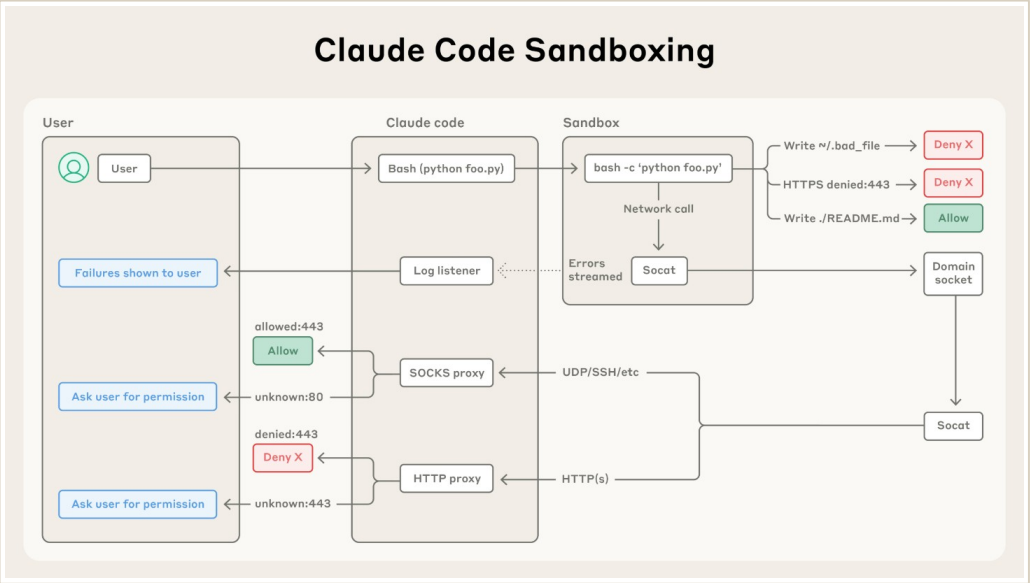
What the evidence says

- DORA calls AI an amplifier: it magnifies an organization's existing strengths and weaknesses.
- Adoption tracks higher delivery throughput and lower delivery stability.
- Google says 75% of its new code is AI-generated and approved by engineers (Apr 2026). The approval is the control.

DORA, State of AI-assisted Software Development (Sep 2025); Stack Overflow Developer Survey 2025; METR (Jul 10, 2025; 2026 redesign has wide intervals); Veracode GenAI Code Security Reports (Jul 2025, Jul 2026); Pichai, Cloud Next blog post (Apr 22, 2026).

Mechanisms that keep coding agents accountable

1. **Context files** (AGENTS.md, CLAUDE.md, copilot-instructions): guidance the agent reads, not enforcement
2. **Specs before code:** Spec Kit (specify, plan, tasks, implement); Kiro (requirements, design, tasks)
3. **Enforcement outside the model:** permission rules, hooks that block tool calls, an OS sandbox for files and network
4. **Review gates:** branch protection, CODEOWNERS, required checks; the requester cannot approve the agent's PR
5. **Evals in CI** for every prompt or model change
6. **Short-lived, scoped credentials;** agents never hold production tokens
7. **Provenance:** signed commits and Co-authored-by trailers record who and what wrote each change



Filesystem and network isolation enforced by the operating system: writes outside the project and unknown hosts are denied or sent to a person.

84% fewer permission prompts with the sandbox on, in Anthropic's internal use

Diagram and figure: Anthropic Engineering, "Claude Code sandboxing" (Oct 20, 2025). AGENTS.md: Agentic AI Foundation, Linux Foundation (Dec 2025). GitHub Spec Kit (Sep 2025); Kiro (Jul 2025); GitHub Copilot cloud agent risks and mitigations.

Exercise: design review for a campus admissions assistant

The case

A university wants an assistant that answers admissions questions from its published rules, looks up an applicant's status, and books appointments with an advisor. It will run on one public cloud and is maintained by a team of three, who use coding agents daily.

1. **Platform.** Choose one cloud and name the managed service for each of the six layers.
2. **Signals.** Define three SLIs and one SLO. What does "HTTP 200 with the wrong content" look like here, and how do you detect it?
3. **Release.** A new prompt version is ready. Which strategy, which gate, and which alarm triggers rollback?
4. **Code.** Two IaC resources you would write, and one policy-as-code rule you would enforce.
5. **Incidents.** For each incident on slide 9, the control in your design that would have stopped it.
6. **Agents.** What can the team's coding agent touch, and what can it never touch?

In pairs, one page. Diagrams may be drawn by hand.